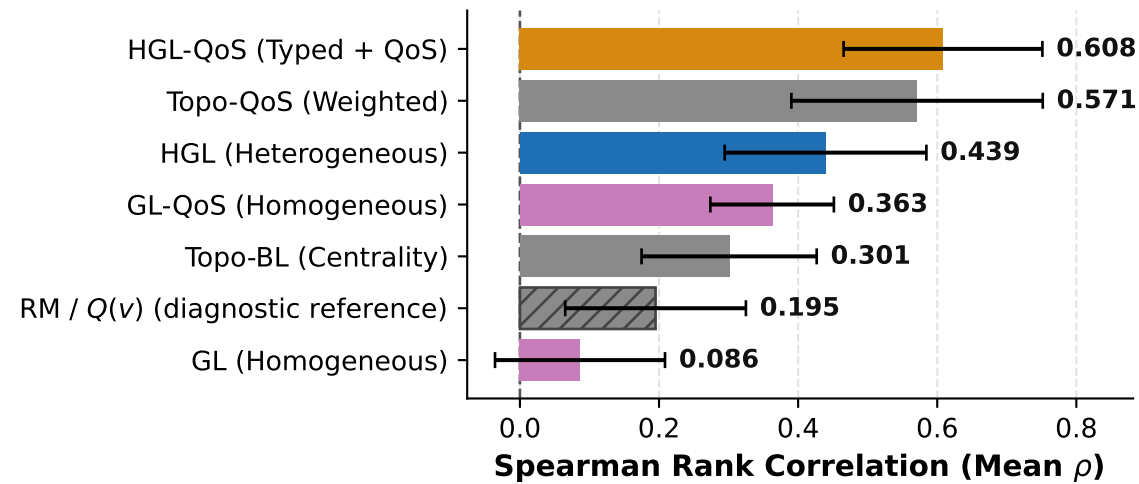
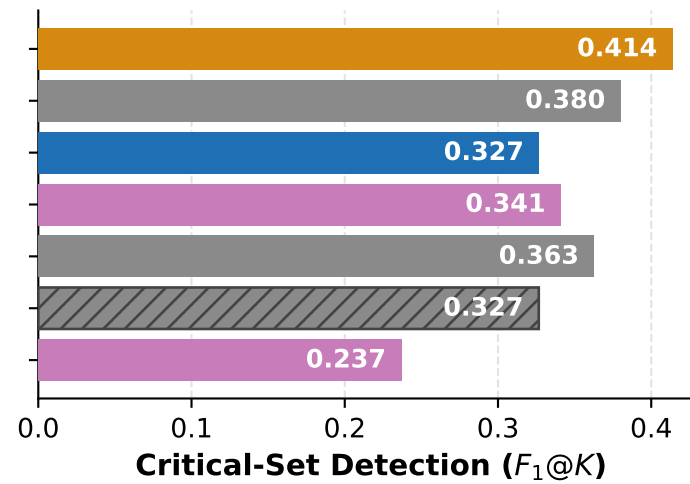


Software-as-a-Graph (SaG) Pre-Deployment Reliability Evaluation — application population

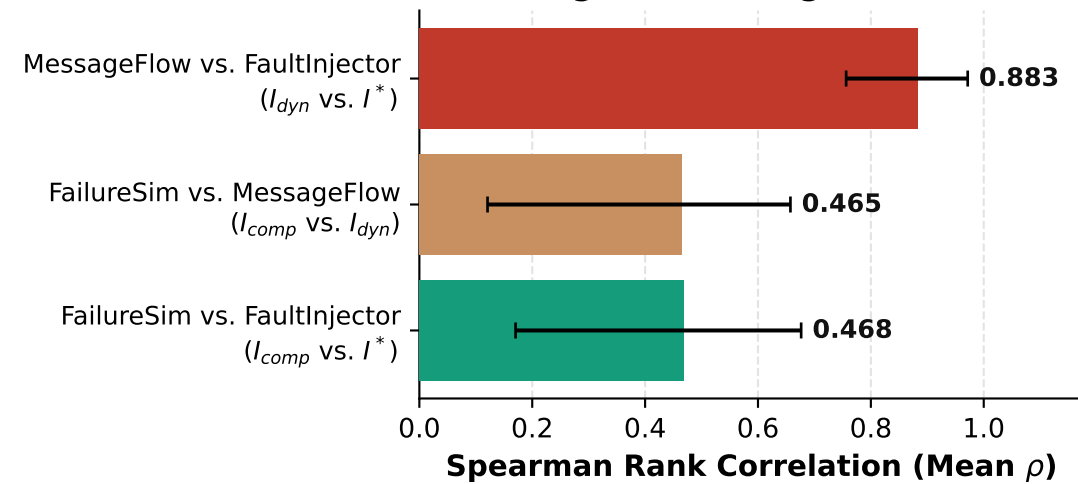
A. Out-of-Distribution Generalization (Leave-One-Scenario-Out ρ)



B. Top-K Critical Component Search ($F_1@K$, $K = \text{top } 20\%$)



C. Simulation Oracle Agreement (Rank Convergence & Range Across Scenarios)



Mean top-K Jaccard overlap: 31.2%–41.9% (chance = 11.1%)
Read against the labeler's own seed-to-seed floor: 44%–100%